ZCAS University

BBF4211 SECURITIES ANALYSIS

DEFERRED EXAMINATION

19 TH JUNE 2026

09:00-12:00

TIME ALLOWED: WRITING – THREE HOURS (READING – 5 MINUTES)

INSTRUCTIONS:

1. Section A: this question is **compulsory** and must be attempted.
2. Sections B: Answer **THREE (3)** questions from this section.
3. This examination paper carries a total of **100 marks**.

Candidates must **not turn this page** until the invigilator tells them to do so.

SECTION A: Question 1 is compulsory and must be attempted

Question 1

Airtel Networks Zambia plc is a LuSE listed company that ranks amongst the top 3 telecom service providers in terms of subscriber numbers. Airtel Networks Zambia plc provides globally admired technologies and services to give Zambia an easy and dependable way to connect to the world. Airtel Networks Zambia plc brings internet to everyone, everywhere. The table below summaries the company's market history over the last 12 years.

Year	Opening Share price	Dividend paid	Closing Share price	Dividend yield %	Capital gain%	Total return%
2014	K26	K1.5	K			
2015	34	1	33			
2016	33	4	32.5			
2017	32.5	4	35			
2018	35	0	35			
2019	35	0	31.48			
2020	31.48	0	29.49			
2021	29.49	2.4	18			
2022	18	5.5	17.25			
2023	17.25	13.2	43.5			
2024	43.5	9.97	43.5			
2025	43.5	15.75	137.73			

Required:

i. Calculate the dividend yield, capital gain yield and total return in percentage. **(20 marks)**

ii. Using the total returns, calculate the following:

1. Mean return. **(3 marks)**
2. Sample variance. **(15 marks)**
3. Standard deviation. **(2 marks)**

(Total: 40 marks)

SECTION B: Attempt any THREE questions in this section

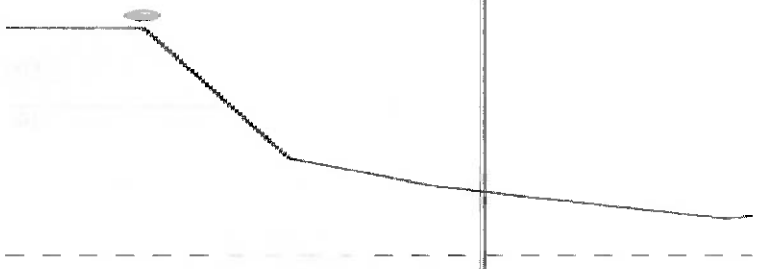
Question 2

On 18th May, 2026, ZAMEFA (the "Board" or "the Company") advised the shareholders that for the year ended 31 March 2026, the basic earnings per share ("EPS") was likely to decline by 161% than that for the year ended 31 March 2025 (K6.62).

The decrease in the EPS is primarily due to:

- The negative effect of the significant strengthening of the Zambian Kwacha against the United States Dollar ("USD") on the revenues which are largely USD denominated;
- A decrease in the overall volumes sold in 2026 as a result of higher copper prices and constrained liquidity affecting some of our major customers in the local market;
- A foreign exchange loss recorded in 2026 as a result of the restatement of the Company's net foreign position compared to a profit in the prior year; and
- The reversal of the plant and equipment impairment in the prior year which was not repeated in the 2026.

The chart below shows the impact of the earnings release upon the share price.



Required:

i. Interpret the share price movement within the context of the Efficient Market Hypothesis, refer to news, speed of reaction and market efficiency in your answer, suggest a reason for the direction of the share price movement. **(6 marks)**

ii. On 18th May, 2026, ZAMEFA plc shares were trading at K54 per share but declined to K51.99 right after the announcement. ZAMEFA has 27,090,100 shares in issue. Calculate the Market capitalization, return on equity (ROE) and PE ratio after the announcement. **(6 marks)**

iii. Conduct E-I-C analysis using ZAMEFA as an example. **(8 marks)**

(Total: 20 marks)

Question 3

In finance, a stock market index or stock index is an index that measures the performance of a stock market, or of a subset of a stock market. The table below contains information about the performance of the LASI in the last 10 years.

Year	LASI in points
2016	4,071.44
2017	5,326.76
2018	5,677.98
2019	4,251.54
2020	3,939.07
2021	6,048.17
2022	7,404.65
2023	11,385.30
2024	16,082.33
2025	27,833.62

Required:

i. Plot the information in the table above and clearly show the trend (peaks and troughs). **(10 marks)**

ii. Discuss ***three*** (3) benefits of the LASI to LuSE investors. **(6 marks)**

iii. List ***four*** (4) commonly used indices in other markets. **(4 marks)**

(Total: 20 marks)

Question 4

Ms Mwaka, a statistics schoolteacher, recently retired and is thinking of making an investment in bonds from her life-long savings of K1 million. She has engaged you based on your expertise in Securities Analysis. She is considering creating an equally weighted bond portfolio to diversify her risks. Ms Mwaka visited the BoZ website and came across the following bond auction results:

TENOR	ISIN NUMBER	COUPON RATE%
2 years	ZM1000007725	14.25
3 years	ZM1000007733	14.5
7 years	ZM1000007683	16
10 years	ZM1000007642	16.6
15 years	ZM1000007659	17.59

Required:

i. Discuss ***three*** (3) risks faced by bondholders. **(6 marks)**

ii. Create a bond ladder and compute the coupons to be received after the first 6 months net of withholding tax and handling fee. **(10 marks)**

iii. Suggest ***two*** (2) other strategies commonly used in the bond market. **(2 marks)**

iv. Explain why the bond market in Zambia is bigger than the stock market. **(2 marks)**

(Total: 20 marks)

Question 5

You are researching on alternative investments and you have come across REIZ-REIT, a listed company on the LuSE dedicated to building long-term wealth through high-yield investments in prime properties. REIZ-REIT focuses on strategic acquisitions and sustainable growth, ensuring strong returns for its investors.

Required:

i. What are alternative investments and what does REIT stand for? **(2 marks)**

ii. Discuss 3 benefits and 3 drawbacks of investing in a REIT. **(12 marks)**

iii. Identify and explain ***three*** (3) other alternative investments. **(6 marks)**

(Total: 20 marks)

END OF EXAMINATION!

FORMULAE SHEET

1. $d_1 = d_0 * (1 + g)$
2. $P_0 = \frac{d_1}{r-g}$
3. $P_0 = \frac{d_0*(1+g_1)^1}{(1+r)^1} + \frac{d_0*(1+g_1)^2}{(1+r)^2} + \frac{d_0*(1+g_1)^3}{(1+r)^3} \ldots\ldots \frac{d_t+P_t}{(1+r)^t}$
4. $P_t = \frac{d_0*(1+g_1)^t * (1+g_2)^1}{r-g_2}$
5. $P_B = C*[\frac{1-\frac{1}{(1+r)^n}}{r}] + FV*(\frac{1}{(1+r)^n})$
6. $YTM = \frac{C+\frac{FV-P}{n}}{0.6P+0.4FV}$
7. $E_R = \sum PiRi$
8. $\sigma 2 = \sum_{i=1}^{n}(R_i - \overline{R})^2(P_i)$
9. $\overline{R} = \frac{(R_1+R_2+\cdots+R_N)}{N}$
10. $S^2 = \frac{\sum_{i=1}^{N}(R_i-\overline{R})^2}{N-1}$
11. $\rho_{x,y} = \frac{\sigma_{xy}}{\sigma_x\sigma_y}$
12. $\beta_a = \frac{\sigma_{am}}{\sigma_m^2}$
13. $Cov(R_aR_b) = \sum(R_a - \overline{R_a})(R_b - \overline{R_b})P$
14. $Cov(R_AR_B) = \frac{\sum(R_A-\overline{R_A})(R_B-\overline{R_B})}{N-1}$
15. 15. $\overline{r_p} = w_x * \overline{r_x} + w_y * \overline{r_y}$
16. $\sigma_p = \sqrt{w_x^2\,\sigma_x^2 + w_y^2\sigma_y^2 + 2\,w_xw_y\sigma_x\sigma_y\rho_{x,y}}$
17. $\bar{r}_i = r_f + \beta_i*(\bar{r}_m - r_f)$
18. $\sigma_i^2 = \beta_i^2\sigma_M^2 + \sigma_\epsilon^2$
19. $TE = \sqrt{\frac{\sum(d-\bar{d})^2}{n-1}}$
20. $IR = \frac{\bar{r}_p-\bar{r}_B}{\sigma_{TE}}$